Physical-domain source page index

H15 NEA 2015: original PDF pages 131, 931, 932, 935, 939, 940

A18 Zhu et al 2018: original PDF pages 3, 5, 7

T20 Zhang et al 2020: original PDF pages 6, 9

Z22 Zhao et al 2022: original PDF pages 6, 9

C23 Cheng et al 2023: original PDF pages 3, 4

X25 Xu et al 2025: original PDF pages 3, 4

Y17 Yue et al 2017: original PDF pages 2, 4

The following pages are unmodified excerpts of user-provided sources.

Table 2.19.2: Summary of the recommended correlations for main thermophysical properties of molten bismuth (Bi)

p ~ 0.1 MPa, except the critical parameters and the saturated vapour pressure

Property, parameter	SI unit	Correlation	Temp. range (K)	Estimated error ±
Melting temperature	K	$T_{M,0} = 544.6$	n/a	0.3
Latent heat of melting	kJ kg^{-1}	$Q_{M,0} = 53.3$	n/a	0.3
Boiling temperature	K	$T_{B,O} = 1\,831$	n/a	6
Latent heat of boiling	kJ kg^{-1}	$Q_{B,0} = 856.2$	n/a	1.9
Critical temperature	K	$T_c = 4\,500$	n/a	500
Critical density	kg m^{-3}	$\rho_c = 2\,800$	n/a	200
Critical pressure	MPa	$p_c = 135$	n/a	15
Saturated vapour pressure	Pa	$p_s = 2.67 \times 10^{10} \cdot \exp(-22\,858/T)$	545-1 831	40%
Surface tension	N m^{-1}	$\sigma = (420.8 - 0.081 \cdot T) \times 10^{-3}$	545-1 400	3%
Density	kg m^{-3}	$\rho = 10\,725 - 1.22 \cdot T$	545-1 300	0.5%
Sound velocity	m s^{-1}	$u_s = 1\,616 + 0.187 \cdot T - 2.2 \times 10^{-4} \cdot T^2$	545-1 800	2%
Bulk modulus	Pa	$B_s = (30.09 - 2.441 \times 10^{-3} \cdot T - 3.913 \times 10^{-6} \cdot T^2) \times 10^9$	545-1 300	5%
Isobaric specific heat	J kg^{-1} K^{-1}	$c_p = 118.2 - 5.934 \times 10^{-3} \cdot T + 7.183 \times 10^6 \cdot T^{-2}$	545-1 300	4%
Dynamic viscosity	Pa s	$\eta = 4.456 \times 10^{-4} \cdot \exp(780/T)$	545-1 300	5%
Electric resistivity	Ω m	$r = (98.96 + 0.0554 \cdot T) \times 10^{-8}$	545-1 420	1.5%
Thermal conductivity	W m^{-1} K^{-1}	$\lambda = 7.34 + 9.5 \times 10^{-3} \cdot T$	545-1 000	(15%)

Table 2.19.3: Summary of the recommended correlations for main thermophysical properties of molten LBE

p ~ 0.1 MPa, except the critical parameters and the saturated vapour pressure

Property, parameter	SI unit	Correlation	Temp. range (K)	Estimated error ±
Melting temperature	K	$T_{M,0} = 398$	n/a	1
Latent heat of melting	kJ kg^{-1}	$Q_{M,0} = 38.6$	n/a	0.3
Boiling temperature	K	$T_{B,O} = 1\,927$	n/a	16
Latent heat of boiling	kJ kg^{-1}	$Q_{B,0} = 856$	n/a	5
Critical temperature	K	$T_c = 4\,800$	n/a	500
Critical density	kg m^{-3}	$\rho_c = 2\,200$	n/a	200
Critical pressure	MPa	$p_c = 160$	n/a	70
Saturated vapour pressure	Pa	$p_s = 1.22 \times 10^{10} \cdot \exp(-22\,552/T)$	670-1 927	60%
Surface tension	N m^{-1}	$\sigma = (448.5 - 0.08 \cdot T) \times 10^{-3}$	420-1 400	3%
Density	kg m^{-3}	$\rho = 11\,065 - 1.293 \cdot T$	400-1 300	0.8%
Sound velocity	m s^{-1}	$u_s = 1\,855 - 0.212 \cdot T$	400-1 100	1%
Bulk modulus	Pa	$B_s = (38.02 - 1.296 \times 10^{-2} \cdot T + 1.32 \times 10^{-6} \cdot T^2) \times 10^9$	400-1 100	3%
Isobaric specific heat	J kg^{-1} K^{-1}	$c_p = 164.8 - 3.94 \times 10^{-2} \cdot T + 1.25 \times 10^{-5} \cdot T^2 - 4.56 \times 10^5 \cdot T^{-2}$	400-1 100	7%
Dynamic viscosity	Pa s	$\eta = 4.94 \times 10^{-4} \cdot \exp(754.1/T)$	400-1 200	8%
Electric resistivity	Ω m	$r = (90.9 + 0.048 \cdot T) \times 10^{-8}$	403-1 100	6%
Thermal conductivity	W m^{-1} K^{-1}	$\lambda = 3.284 + 1.617 \times 10^{-2} \cdot T - 2.305 \times 10^{-6} \cdot T^2$	403-1 100	(15%)

15.3 Pressure loss correlations used in design and analysis models

The thermal-hydraulic analysis of a complex cooling system is usually performed by means of system codes with suitable lamped models in order to describe all relevant physical phenomena and components. The increased calculation power has made possible the use of detailed local calculation by CFD codes for the most complex phenomena and components. Both system and CFD codes have been applied by different participants in LACANES. The different results are compared and analysed in the following sections so as to evaluate the code predictions and provide recommendations for the thermal-hydraulic modelling of the HLM loop.

15.3.1 *Design and analysis codes used in benchmarking*

The system codes applied in LACANES benchmarking Phase I were HETRAF, RELAP5, LegoPST, MARS-LBE and TRACE. CFD codes were CFX® and Star-CD®.

Table 15.3.1: All codes used in the LACANES benchmark programme

Code	Type	Reference
HETRAF	System code	Cheng (1994)
RELAP5	System code	INL (2001)
LegoPST	System code	Casamassima (2008)
MARS-LBE	System code	KAERI (2006)
TRACE	System code	US NRC (2007a, 2007b)
CFX®	Computational fluid dynamics	ANSYS (2009)
Star-CD®	Computational fluid dynamics	CD-adapco (2007)

Each system code has a module that calculates pressure losses by defining the friction loss coefficients and form loss coefficients with Eq. (15.3.1):

$$\Delta P_{total} = \frac{1}{2}\rho \sum_{i} V^2_{i(average)} \left(f\frac{L}{D} + K \right)_i \tag{15.1}$$

where P, i, ρ, V, f, L, D and K are pressure, the number of components, fluid density, average flow velocity, friction factor, length, the diameter of a component and the form loss coefficient, respectively.

The last term in the right hand side of Eq. (15.3.1) is defined as:

$$\text{Pressure loss coefficient} = \left(f\frac{L}{D} + K \right) \tag{15.2}$$

where $f(L/D)$ is the friction loss coefficient for a component with no change in cross-sectional dimensions.

Pressure loss coefficient for each geometry was calculated by the correlations located in the internal structure of code that was induced by the hydraulic-resistance handbooks. If there are no internal correlations or parameters for pressure loss coefficients in the system codes, it should be defined by users from the hydraulic-resistance handbooks. In the following sections, all correlations or parameters employed by participants to predict the pressure losses of HELIOS loops are introduced.

15.3.2 *Pipe wall friction loss correlations*

For the calculation of the distributed pressure loss all codes utilise a relation of Eq. (15.1), where the friction factor is evaluated with different correlations. These correlations are reported in Table 15.3.2 where information is provided on the range of applicability and literature references.

Table 15.3.2: Formulas for calculation of friction loss coefficients

		Correlations	Applicable range	Reference
Laminar flow		$f = \frac{64}{Re}$ (Hagen-Poiseuille law)	Up to $Re \approx 2\,000$	White (1986)
Transient or turbulent flow	1	$f_{L,T} = \left(3.75 - \frac{8250}{Re}\right)\left(f_{T,3000} - f_{L,2200}\right) + f_{L,2200}$ $f_{L,2200}$: Laminar friction factor at Re = 2 200 $f_{T,3000}$: Turbulent friction factor at Re = 3 000	2 200 < Re < 3 000	Colerbrook (1939) Zigrang (1985)
		$\frac{1}{\sqrt{f_T}} = -2\log_{10}\left\{\frac{\varepsilon}{3.7D} + \frac{2.51}{Re}\left[1.14 - 2\log_{10}\left(\frac{\varepsilon}{D} - \frac{21.25}{Re^{0.9}}\right)\right]\right\}$	Re > 3 000	
	2	$\frac{1}{\sqrt{f}} = -2\log_{10}\left\{\frac{\varepsilon}{3.7D} + \frac{2.51}{Re\sqrt{f}}\right\}$ (Colebrook's interpolation formula)	Re > 2 000	Idelchik (1994)
	3	$f = 0.11\left(\frac{68}{Re} + \frac{\varepsilon}{D}\right)^{0.25}$ (Altshul's approximate formula)	Re > 2 090·$(\varepsilon/D)^{0.635}$	Idelchik (1994)
	4	$f = \frac{1}{\left[2\log_{10}\left(\frac{2.51}{Re\sqrt{f}}\right) + \frac{\varepsilon}{3.7D}\right]^2}$	Re > 2 090·$(\varepsilon/D)^{0.635}$	Idelchik (1994)
	5	$f = \frac{0.3164}{Re^{0.25}}$ (Blasius)	4 000 < Re < 10^5	Idelchik (1994)
	6	$f = 8\cdot\left[\left(\frac{8}{Re}\right)^{12} + \frac{1}{(A+B)^{1.5}}\right]^{1/12}$ $A = \left\{2.457\cdot\ln\left[\left(\frac{7}{Re}\right)^{0.9} + 0.27\cdot\left(\frac{\varepsilon}{D}\right)\right]\right\}^{16}$ $B = \left(\frac{37530}{Re}\right)^{16}$ (Churchill's formula)	All ranges	Idelchik (1994)
	7	$f = \left(1.8\log_{10}\frac{8.25}{\frac{56}{Re} + \frac{\varepsilon}{D}}\right)^{-2}$	For turbulent flow	Slissky (1983)

f, r, D, Re_D and ε are friction factor, radius of pipe, diameter of pipe, Reynolds number and absolute surface roughness, respectively.

developed a correlation of grid spacer in a fuel bundle by many sets of grid spacers' experimental data that pronounced the effect of the blocking area ratio ($A_{grid\text{-}spacer}/A_{flow}$) as shown in the formulation of the Table 15.3.4.

Figure 15.3.4: Configuration of spacer

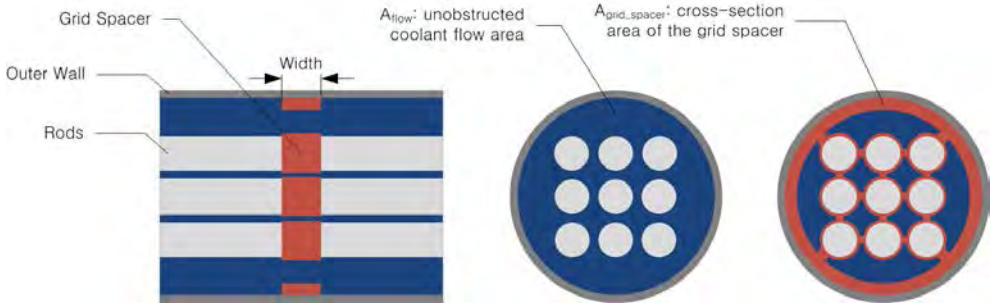


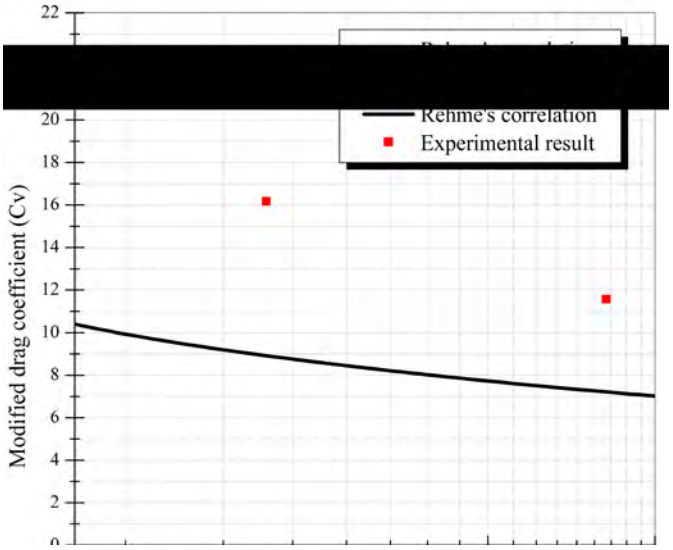
Table 15.3.4: Formulas for calculation of form loss coefficients in spacers

	Correlations	Reference
1	$K = C_v \left(\frac{A_{grid\text{-}spacer}}{A_{flow}} \right)^2$ $C_v = 3.5 + \frac{73.5}{Re^{0.264}} + \frac{2.79 \cdot 10^{10}}{Re^{2.79}}$ (C_v: modified drag coefficient)	Rehme (1973)

Where $A_{grid\text{-}spacer}$ and A_{flow} are the cross-section area of the grid spacer and the unobstructed coolant flow area, respectively.

In Rehme's correlation the predicted pressure losses of grid spacer depend on the area ratio of obstruction and flow ($A_{grid\text{-}spacer}/A_{flow}$), and on modified drag coefficient which is in turn determined as function of Reynolds number. However, experimental results from the HELIOS loop showed a discrepancy from those predictions with Rehme's correlation, as shown in Figure 15.3.5.

Figure 15.3.5: Comparison of grid spacer correlation (Rehme's correlation) with experimental result



The most important feature of the computed pressure distribution is its step profile with near uniform pressure in the subchannels and large pressure jumps across each of the spacers. Pressure overshoots are predicted in the spacer regions because spacers have not only sudden expansions but also sudden contractions. As the flow area is suddenly contracted (to one-half of the flow area at the spacers' entrance), the pressure is sharply decreased. Then, pressure is recovered immediately after because flow area is suddenly expanded, having twice the flow area at the spacers' exit (Batta, 2010).

Based on the system code modelling of pressure losses in the mock-up core region, a substantial discrepancy from experimental data was found. CFD simulations for the mock-up core also showed that the pressure drop is dominated by spacer pressure losses. Each of the three spacers results in substantially different pressure losses due to different levels of flow development. Measured total pressure loss of the mock-up core (52.5 kPa) matches CFD results (46 kPa) very well.

15.4 Best practice guidelines

Based on findings from LACANES benchmarking, a set of best practice guidelines has been recommended in Table 15.4.1 for the prediction of pressure losses in LACANES. In the CFD simulations that provided accurate predictions, all prediction results were always in good agreement with experimental data within ±15% in pressure drop. When geometry is complicated and there is no available experimental data, CFD simulation can be recommended as the best practice.

Table 15.4.1: Recommended correlations

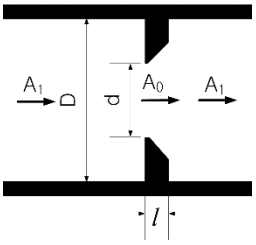
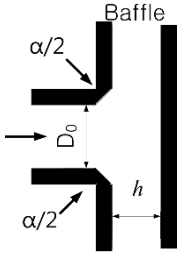
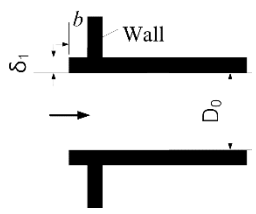
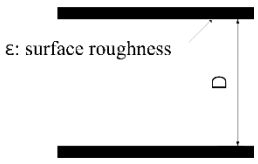
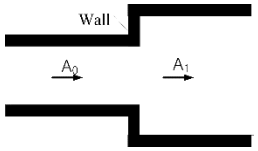
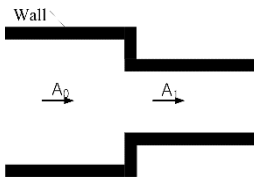
<table>
<tr><th>Components</th><th>Geometry</th><th>Recommended correlations</th><th>Reference</th></tr>
<tr><td>Orifice</td><td> 
</td><td>$K = \left(1 + 0.707\sqrt{1 - \frac{F_0}{F_1}} - \frac{F_0}{F_1}\right)^2 \left(\frac{F_1}{F_0}\right)^2$
$Re > 10^5$</td><td>Idelchik (1994)</td></tr>
<tr><td>Discharge into a vessel</td><td> 
</td><td>
<table>
<tr><th rowspan="2">α deg</th><th colspan="10">h/D0</th></tr>
<tr><th>0.10</th><th>0.15</th><th>0.20</th><th>0.25</th><th>0.30</th><th>0.40</th><th>0.50</th><th>0.60</th><th>0.70</th><th>1.0</th></tr>
<tr><td>0</td><td>–</td><td>–</td><td>–</td><td>–</td><td>–</td><td>–</td><td>1.37</td><td>1.20</td><td>1.11</td><td>1.00</td></tr>
<tr><td>15</td><td>–</td><td>–</td><td>–</td><td>1.50</td><td>1.06</td><td>0.72</td><td>0.61</td><td>0.59</td><td>0.58</td><td>0.58</td></tr>
<tr><td>30</td><td>–</td><td>–</td><td>1.23</td><td>0.79</td><td>0.66</td><td>0.64</td><td>0.66</td><td>0.66</td><td>0.67</td><td>0.67</td></tr>
<tr><td>45</td><td>–</td><td>1.50</td><td>0.85</td><td>0.73</td><td>0.75</td><td>0.79</td><td>0.81</td><td>0.82</td><td>0.82</td><td>0.82</td></tr>
<tr><td>60</td><td>–</td><td>0.98</td><td>0.76</td><td>0.80</td><td>0.90</td><td>0.96</td><td>1.00</td><td>1.01</td><td>1.02</td><td>1.02</td></tr>
<tr><td>90</td><td>1.50</td><td>0.72</td><td>0.74</td><td>0.83</td><td>0.89</td><td>0.94</td><td>0.96</td><td>0.98</td><td>1.00</td><td>1.00</td></tr>
</table>
</td><td>Idelchik (1994)</td></tr>
<tr><td>Entrance in tubes</td><td> 
</td><td>
<table>
<tr><th rowspan="2">δ1/D0</th><th colspan="11">b/D0</th></tr>
<tr><th>0</th><th>0.002</th><th>0.005</th><th>0.010</th><th>0.020</th><th>0.050</th><th>0.100</th><th>0.200</th><th>0.300</th><th>0.500</th><th>∞</th></tr>
<tr><td>0</td><td>0.50</td><td>0.57</td><td>0.63</td><td>0.68</td><td>0.73</td><td>0.80</td><td>0.86</td><td>0.92</td><td>0.97</td><td>1.00</td><td>1.00</td></tr>
<tr><td>0.004</td><td>0.50</td><td>0.54</td><td>0.58</td><td>0.63</td><td>0.67</td><td>0.74</td><td>0.80</td><td>0.86</td><td>0.90</td><td>0.94</td><td>0.94</td></tr>
<tr><td>0.008</td><td>0.50</td><td>0.53</td><td>0.55</td><td>0.58</td><td>0.62</td><td>0.68</td><td>0.74</td><td>0.81</td><td>0.85</td><td>0.88</td><td>0.88</td></tr>
<tr><td>0.012</td><td>0.50</td><td>0.52</td><td>0.53</td><td>0.55</td><td>0.58</td><td>0.63</td><td>0.68</td><td>0.75</td><td>0.79</td><td>0.83</td><td>0.83</td></tr>
<tr><td>0.016</td><td>0.50</td><td>0.51</td><td>0.51</td><td>0.53</td><td>0.55</td><td>0.58</td><td>0.64</td><td>0.70</td><td>0.74</td><td>0.77</td><td>0.77</td></tr>
<tr><td>0.020</td><td>0.50</td><td>0.51</td><td>0.51</td><td>0.52</td><td>0.53</td><td>0.55</td><td>0.60</td><td>0.66</td><td>0.69</td><td>0.72</td><td>0.72</td></tr>
<tr><td>0.024</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.51</td><td>0.52</td><td>0.53</td><td>0.58</td><td>0.62</td><td>0.65</td><td>0.68</td><td>0.68</td></tr>
<tr><td>0.030</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.52</td><td>0.52</td><td>0.52</td><td>0.54</td><td>0.57</td><td>0.59</td><td>0.61</td><td>0.61</td></tr>
<tr><td>0.040</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.51</td><td>0.51</td><td>0.51</td><td>0.51</td><td>0.52</td><td>0.52</td><td>0.54</td><td>0.54</td></tr>
<tr><td>0.050</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td></tr>
<tr><td>∞</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td><td>0.50</td></tr>
</table>
</td><td>Idelchik (1994)</td></tr>
</table>

Table 15.4.1: Recommended correlations (*cont.*)

Components	Geometry	Recommended correlations	Reference
Straight pipe		$f = 8 \cdot \left[\left(\frac{8}{Re} \right)^{12} + \frac{1}{(A+B)^{1.5}} \right]^{1/12}$ $A = \left\{ 2.457 \cdot \ln \left[\left(\frac{7}{Re} \right)^{0.9} + 0.27 \cdot \left(\frac{\varepsilon}{D} \right) \right] \right\}^{16}$ $B = \left(\frac{37530}{Re} \right)^{16}$	Idelchik (1994)
Sudden changes		$K_{SE} = \left(1 - \frac{F_0}{F_1} \right)^2$ for sudden expansion	Idelchik (1994)
		$K_{SC} = 0.5 - 0.7 \cdot \left(\frac{F_1}{F_0} \right) + 0.2 \left(\frac{F_1}{F_0} \right)^2$ for sudden contraction	Massey (1968)

15.5 Summary and recommendations for future work

Guidelines for the prediction of pressure losses were obtained based on a comparison between predictions calculated by system codes or CFD codes and experimental data from HELIOS measurements. The comparison included both simple geometries such as pipe and complicated ones such as the mock-up core region. For simple geometries existing correlations were found adequate for use in design and analysis. However, experimental data or CFD modelling is needed for complicated geometries.

These guidelines can be recommended only for the range of Reynolds number of the HELIOS experiments, i.e. $10^4 < Re_D < 2 \cdot 10^5$.

Future benchmarking with natural circulation will be used to verify that the best practice guidelines are applicable not only for forced convection but also for natural circulation, which is becoming increasingly important for passive safety assurance.

钢，加热丝和外壳之间填充氧化镁粉，热电偶贴附在外壳的内壁上。实验段外套管为内径 26 mm、壁厚 3 mm 的不锈钢管，与电加热元件组成内径 20 mm、外径 26 mm、长 1 400 mm 的环形实验段。电加热元件与上、下联箱采用焊接的方式连接。

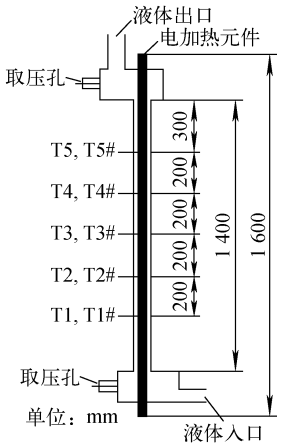


图 2　实验段示意图

Fig. 2　Schematic diagram of experimental annular channel

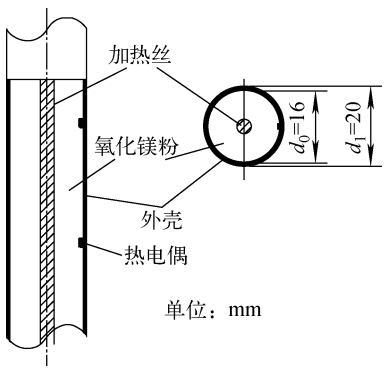


图 3　电加热元件示意图

Fig. 3　Schematic diagram of electric heating element

2　实验数据处理

2.1　铅铋合金热物性

相同压力、温度条件下，液态铅铋合金的热物性和水有很大区别。世界经济合作组织核能署（OECD/NEA，Organisation for Economic Cooperation and Development/Nuclear Energy Agency）在 2007 年出版的铅铋手册中对前人的一些实验结果进行了整理和分析，总结出了一套铅铋合金物性的计算公式[13]，其中铅铋合金的密度 ρ_{LBE}、比定压热容 $c_{p,\mathrm{LBE}}$、动力黏度 η_{LBE} 和热导率 λ_{LBE} 如下所示：

$$\rho_{\mathrm{LBE}} = 11\,096 - 1.323\,6T \tag{1}$$

$$c_{p,\mathrm{LBE}} = 159 - 2.72\times10^{-2}T + 7.12\times10^{-6}T^2 \tag{2}$$

$$\eta_{\mathrm{LBE}} = 4.94\times10^{-4}\exp(757.1/T) \tag{3}$$

$$\lambda_{\mathrm{LBE}} = 3.61 + 1.517\times10^{-2}T \tag{4}$$

式中，T 为流体的平均温度。因此，可根据实验数据得到实验段流体的平均温度，进而求得液态铅铋合金的密度、比定压热容、动力黏度及热导率。

2.2　流动特性实验数据处理

液态铅铋合金在环形通道内总流动压降 Δp 为：

$$\Delta p = \Delta p_{\mathrm{el}} + \Delta p_{\mathrm{a}} + \Delta p_{\mathrm{f}} + \Delta p_{\mathrm{c}} \tag{5}$$

其中：Δp_{el} 为重位压降，Pa；Δp_{a} 为加速压降，Pa；Δp_{f} 为摩擦压降，Pa；Δp_{c} 为局部阻力压降，Pa。计算公式如下：

$$\Delta p_{\mathrm{el}} = \bar{\rho} g \Delta h \tag{6}$$

$$\Delta p_{\mathrm{a}} = G^2\left(\frac{1}{\rho_2} - \frac{1}{\rho_1}\right) \tag{7}$$

$$\Delta p_{\mathrm{c}} = (\xi_1 + \xi_2)\frac{\bar{\rho}u^2}{2} \tag{8}$$

$$\xi_1 = (1 - A_0/A_1)^2 \tag{9}$$

$$\xi_2 = 0.5[1 - (A_0/A_1)^2] \tag{10}$$

$$u = q_{\mathrm{m}}/\bar{\rho}A_0 \tag{11}$$

$$f = \frac{2D_{\mathrm{e}}\Delta p_{\mathrm{f}}}{\bar{\rho}Lu^2} \tag{12}$$

$$D_{\mathrm{e}} = d_2 - d_1 \tag{13}$$

$$Re = uD_{\mathrm{e}}\bar{\rho}/\eta \tag{14}$$

式中：$\bar{\rho}$ 为实验段内流体平均密度，kg/m^3；Δh 为取压孔之间的高度，m；G 为质量流密度，kg/(m^2·s)；ρ_1 为实验段入口流体密度，kg/m^3；ρ_2 为实验段出口流体密度，kg/m^3；ξ_1 为突扩行阻系数；ξ_2 为突缩行阻系数；A_0 为环形通道截面积，m^2；A_1 为上联箱截面积，m^2；A_2 为下联箱截面积，m^2；u 为实验段流体速度，m/s；q_{m} 为铅铋合金质量流量，kg/s；f 为摩擦系数；D_{e} 为实验段当量直径，m；L 为实验段长度，m；d_1 为环形通道内径，m；d_2 为环形通道外径，m；Re 为Reynolds 数。

由于进出口流体密度变化很小，因此加速压降可忽略不计。由式(5)～(14)可得出在不同 Re 下 f 的变化趋势，即 Re-f 关系曲线。本

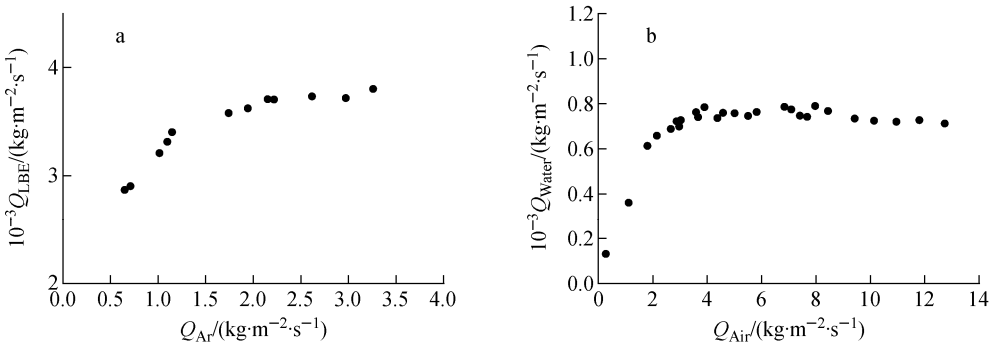


图 4　液体质量流速随注气速度的变化

Fig. 4　Change of liquid mass flow rate under different gas-injection velocities

后,液态金属和水的质量流速均不再增大。但注气速度达到 2.2 kg/(m^2 · s)(0.6 kg/h)后,液态金属的质量流速不再增大,而注气速度达到4 kg/(m^2 · s)(6 kg/h)后,水的质量流速才不再增大。本实验给出的是注气速度对液态金属质量流速的提升效果,文献中给出的是浸没比为 0.74 时,注气速度对水质量流速的影响。依据文献中给出的浸没比越大,水的质量流速越大的结论,注气速度对液态金属质量流速的提升效果应大于浸没比为 0.74 时的注气速度对水质量流速的提升,但由于液态金属回路中回路管道更长,摩擦阻力更大,因此,液态金属质量流速的增强只是略强于浸没比为 0.74 时的水的质量流速提升量。

2) 流动特性分析

随 Re 的增大,流动可分为层流区($Re\leqslant 2\,000$)、过渡区($2\,000<Re\leqslant 4\,000$)以及湍流区($Re>4\,000$)。本实验在湍流区的范围内,研究了液态铅铋合金在环形通道内流动的摩擦压降和 f。

图 5 为不同 Re 下摩擦压降的变化。从图 5可看出,随 Re 的增加,环形通道内液态铅铋合金的摩擦压降也增加。这是因为 Re 增大即为流动速度增加,随流速的增加,重位压降保持不变,局部阻力压降增大,但局部压降增大的幅度相比于总压降可忽略不计。因此,总压降的增大主要是由摩擦压降引起的,即流速增加,流体的摩擦压降增大。

在光滑圆管湍流区,水等常规牛顿流体的 f 常用布拉休斯(Blasius)公式计算:

$$f = 0.316\,4/Re^{0.25} \tag{25}$$

对于环形通道内液态铅铋合金的 f 拟合采用如下关系式:

$$f = c_1/Re^{c_2} \tag{26}$$

其中,c_1、c_2 为需要拟合的参数。

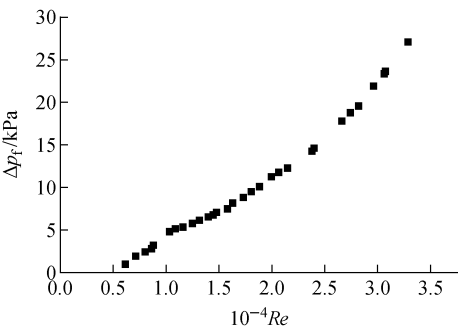


图 5　摩擦压降随 Re 的变化

Fig. 5　Change of frictional pressure drop under different Re

结合实验数据拟合出 f 与 Re 的关系式为:

$$f = 0.685/Re^{0.3} \tag{27}$$

式(27)适用范围为:$5\,000<Re<35\,000$。

f 的拟合曲线与实验结果的对比如图 6 所示。由图 6 可看出,液态铅铋合金的 f 随 Re 的增加而降低,实验数据和拟合曲线的相对误差在 10%以内。图 6 中还给出了由布拉休斯公式(式(25))计算得到的 f 曲线,由图可见,由实验得到的 f 大于相同 Re 下由布拉休斯公式计算得到的,这是因为布拉休斯公式适用于光滑圆管,本实验是在狭小的环形通道($d_2/d_1=1.3$)中进行的,在相同 Re、水力直径下的狭小环形通道与圆管相比,环形通道的截面润湿周长更大,液体与壁面接触的面积就更大。

1) 气泡泵注气速度小于 2.2 kg/(m^2 · s)(0.6 kg/h)时,液态铅铋合金的质量流速随注气速度的增加也会显著提高,当注气速度达到 2.2 kg/(m^2 · s)(0.6 kg/h)时,液态铅铋合金流动趋于稳定,液态铅铋合金的质量流速基本上不再随注气速度的增加而改变。将本实验结果与注气速度对水质量流速影响的实验结果进行对比,发现两者具有相同的变化趋势,为后续研究不同浸没比下注气速度对液态金属质量流速的影响提供参考。

2) 环形通道内湍流流动状态下,液态铅铋合金流动的 f 大于相同 Re、相同水力直径下由布拉休斯公式计算得到的 f,由实验数据拟合得到的 f 关系式为 $f=\frac{0.685}{Re^{0.3}}$(5 000<Re<35 000),计算值与实验数据相对误差在 10%内。

3) 在 100<Pe<450 范围内,对于液态铅铋合金的对流换热特性拟合得到的关系式为 $Nu=(4.05+0.021Pe^{0.8})(d_2/d_1)^{0.3}$,计算值与实验数据相对误差在±7%内。

参考文献:

[1] LITFIN K, BATTA A, CLASS A G, et al. Investigation on heavy liquid metal cooling of ADS fuel pin assemblies[J]. Journal of Nuclear Materials, 2011, 415(3): 425-432.

[2] ZUO J, TIAN W, CHEN R, et al. Two-dimensional numerical simulation of single bubble rising behavior in liquid metal using moving particle semi-implicit method[J]. Progress in Nuclear Energy, 2013, 64(12): 31-40.

[3] BENAMATI G, FOLETTI C, FORGIONE N, et al. Experimental study on gas-injection enhanced circulation performed with the CIRCE facility[J]. Nuclear Engineering and Design, 2007, 237(7): 768-777.

[4] TARANTINO M, BERNARDI D, COCCOLUTO G, et al. Natural and gas enhanced circulation tests in the NACIE heavy liquid metal loop[C]// International Conference on Nuclear Engineering. [S. l.]: [s. n.], 2010: 933-942.

[5] SUZUKI T, TOBITA Y, KONDO S, et al. Analysis of gas-liquid metal two-phase flows using a reactor safety analysis code SIMMER-Ⅲ[J]. Nuclear Engineering and Design, 2003, 220(3): 207-223.

[6] SAITO Y, MISHIMA K, TOBITA Y, et al. Velocity field measurement in gas-liquid metal two-phase flow with use of PIV and neutron radiography techniques[J]. Applied Radiation and Isotopes, 2004, 61(4): 683-691.

[7] SAITO Y, MISHIMA K, TOBITA Y, et al. Application of high frame-rate neutron radiography to liquid-metal two-phase flow research[J]. Nuclear Instruments and Methods in Physics Research, 2005, 542(1): 168-174.

[8] AMBROSINI W, FORASASSI G, FORGIONE N, et al. Experimental study on combined natural and gas-injection enhanced circulation[J]. Nuclear Engineering and Design, 2005, 235(10-12): 1 179-1 188.

[9] 仇子铖,秋穗正,巫英伟,等. 环形通道内液态金属钠流动换热特性实验研究[J]. 原子能科学技术,2015,49(5):795-800.
QIU Zicheng, QIU Suizheng, WU Yingwei, et al. Experimental research on heat transfer characteristic of liquid sodium flowing in annulus[J]. Atomic Energy Science and Technology, 2015, 49(5): 795-800(in Chinese).

[10] TALER D. Heat transfer in turbulent tube flow of liquid metals[J]. Procedia Engineering, 2016, 157: 148-157.

[11] PACIO J, MAROCCO L, WETZEL T. Review of data and correlations for turbulent forced convective heat transfer of liquid metals in pipes[J]. Heat and Mass Transfer, 2015, 51(2): 153-164.

[12] JAEGER W. Heat transfer to liquid metals with empirical models for turbulent forced convection in various geometries[J]. Nuclear Engineering and Design, 2017, 319: 12-27.

[13] Handbook on lead-bismuth eutectic alloy and lead properties, materials compatibility, thermal-hydraulics and technologies[R]. [S. l.]: Organisation for Economic Cooperation and Development, Nuclear Energy Agency, 2007.

[14] MOFFAT R J. Describing the uncertainties in experimental results[J]. Experimental Thermal and Fluid Science, 1988, 1(1): 3-17.

[15] KASSAB S Z, KANDIL H A, WARDA H A, et al. Air-lift pumps characteristics under two-phase flow conditions[J]. International Journal of Heat and Fluid Flow, 2009, 30(1): 88-98.

Table 3
Experimental conditions.

Variables	Value
Inner diameter of the test section	20 mm, 35 mm
Inlet temperature	300 °C, 400 °C
Mass flow rate	0 − 12 kg/s
Heat flux	20, 000 − 50, 000 W/m^2

Table 4
The accuracy of the measurement.

Instrument	Uncertainty/%
Pressure sensor	± 0.075
EMF	± 2
Voltmeter	± 0.5
Ampere meter	± 0.2
Thermo-couple	± 0.75
Diameter measurement	± 0.01
Length measurement	± 0.05
Data acquisition system	± 0.02

$$\frac{\delta R}{R} = \sqrt{\left(\frac{\delta e_1}{e_1}\right)^2 + \left(\frac{\delta e_2}{e_2}\right)^2 + ... + \left(\frac{\delta e_n}{e_n}\right)^2} \tag{16}$$

Therefore, the uncertainty of a measuring parameter in this experiment is:

$$\frac{\delta R}{R} = \sqrt{\left(\frac{\delta R_{dau}}{R_{dau}}\right)^2 + \left(\frac{\delta R_{miu}}{R_{miu}}\right)^2} \tag{17}$$

Based on Table 4, the uncertainty of each parameter, including directly measured parameters and derived parameters, used in the analysis is calculated, as shown in Table 6.

4. Experimental results and discussion

4.1. Frictional resistance coefficient

The pressure drop measured by the differential pressure sensors in the experiment is composed of gravity pressure drop, accelerated pressure drop and frictional pressure drop, as shown below:

$$\Delta P = \Delta P_{grav} + \Delta P_{acc} + \Delta P_f \tag{18}$$

The calculation formulas of the terms are as follows:

$$\Delta P_{grav} = \bar{\rho} g \Delta H \tag{19}$$

$$\Delta P_{acc} = G^2\left(\frac{1}{\rho_2} - \frac{1}{\rho_1}\right) \tag{20}$$

$$\Delta P_f = f \cdot \frac{1}{2}\rho u^2 \cdot \frac{l}{D} \tag{21}$$

As mentioned in Section 3.1, the temperatures at the inlet and outlet are set to equal, so the density at the inlet and outlet is equal. As a result, ΔP_{acc} in Eq. (18) is eliminated. Finally, the frictional resistance coefficient is calculated using Eq. (21).

The frictional resistance coefficient of liquid metals is similar to that of water according to the former research. Therefore, the two coefficients are also compared in this paper. The frictional resistance coefficient correlation of water is based on the research of Nikuradse [31,32], in which the relation between the *Re* and frictional resistance coefficient in pipes is established considering the influence of roughness. As the test section in this experiment could be treated as new seamless steel pipes, the roughness used in the Nikuradse experiment is chosen as 0.15 mm according to reference [33]. The range of *Re* in this experiment is $3.8 \times 10^4 < \mathrm{Re} < 1.8 \times 10^5$, and according to the criterion of *Re* (Eq. (22)), the correlation for the turbulent rough pipe transition range should be applied, as shown in Eq. (23).

$$26.98\left(\frac{d}{\Delta}\right)^{\frac{8}{7}} < Re < 4160\left(\frac{d}{2\Delta}\right)^{0.85} \tag{22}$$

$$f = 0.11\left(\frac{\Delta}{d} + \frac{68}{\mathrm{Re}}\right)^{0.25} \tag{23}$$

The measurement on the frictional resistance coefficient is carried out at the inlet temperature of 350 °C within the test section with the I.D. of 20 mm, in order to get a relatively larger pressure drop and to decrease the experiment error. The comparison between the frictional resistance coefficient of LBE obtained in this experiment and the correlation of the Nikuradse experiment is shown in Fig. 6. It can be seen from the figure that the results in this experiment show the same variation tendency with the correlation of Nikuradse while the value is 1.5 times larger. The deviation mainly comes from several aspects. Firstly, LBE has twice the viscosity of water, which leads to a higher frictional pressure drop. The difference between the physical properties of LBE and water is the main reason for the deviation of the results. On the other hand, the impurity in LBE, including oxide and the debris of the pipes, may have an influence on the physical property of LBE and enlarge the flow resistance on the contact surface. In addition, the roughness of the test section is also an important factor influencing the frictional resistance coefficient. The roughness in this paper is obtained by the recommended value without actual measurement and further investigation should focus on this aspect. The correlation of the frictional resistance coefficient of LBE fitted by the experimental data is shown in Eq. (24).

$$f = \frac{0.1581}{Re^{0.13}} \tag{24}$$

4.2. Heat transfer coefficient

The heat transfer coefficient is obtained based on the following equation:

$$q = h(T_w - T_f) \tag{25}$$

The heat flux q is calculated by Eq. (13). The wall temperature T_w is calculated using Eq. (11) and the fluid temperature T_f is directly measured.

There are four pairs of TCs evenly arranged along the axial direction of the test section for the measurement of T_w and T_f, as shown in Fig. 4. The data obtained by the $T_w - 4$ and $T_f - 4$are adopted in the analysis to make sure that the flow has been fully-developed.

Table 5
Physical properties of LBE.

Property	Relation	Unit	Uncertainty
Density	$\rho = 11096 - 1.3236 \cdot T$	kg·m^{-3}	0.8%
Heat capacity	$c_p = 159 - 2.72 \times 10^{-2}T + 7.12 \times 10^{-6}T^2$	J·kg^{-1}·K^{-1}	5.0%
Thermal conductivity	$\kappa = 3.61 + 1.517 \times 10^{-2}T - 1.741 \times 10^{-6}T^2$	W·m^{-1}·K^{-1}	5.0%
Dynamic viscosity	$\mu = 4.94 \times 10^{-4} \times \exp(754.1/T)$	Pa·s	6.0%

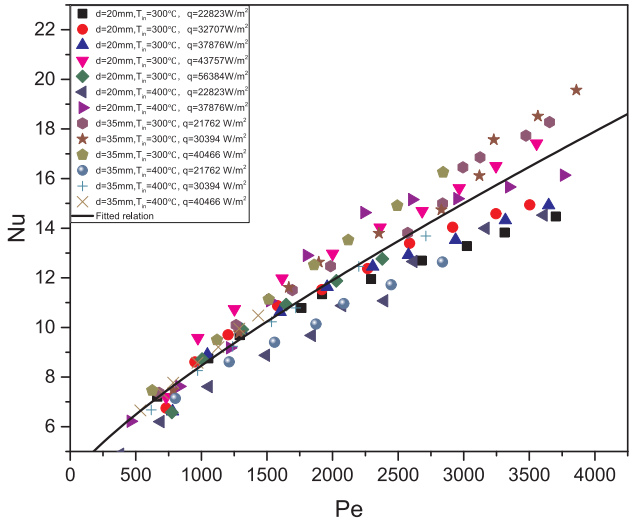


Fig. 10. The relation of *Nu* and *Pe*.

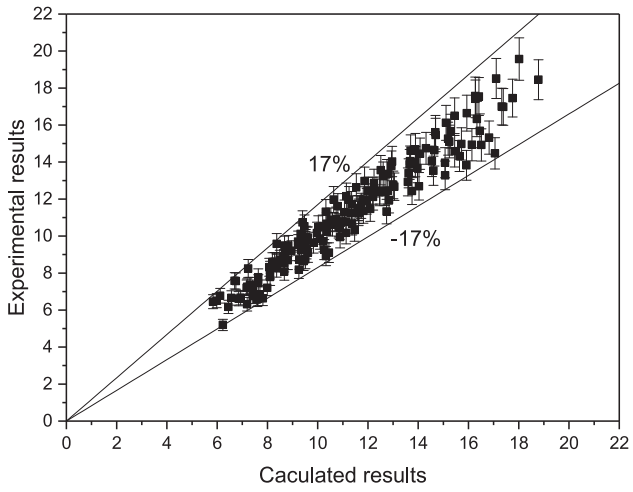


Fig. 11. The relative deviation between experimental data and fitted correlation.

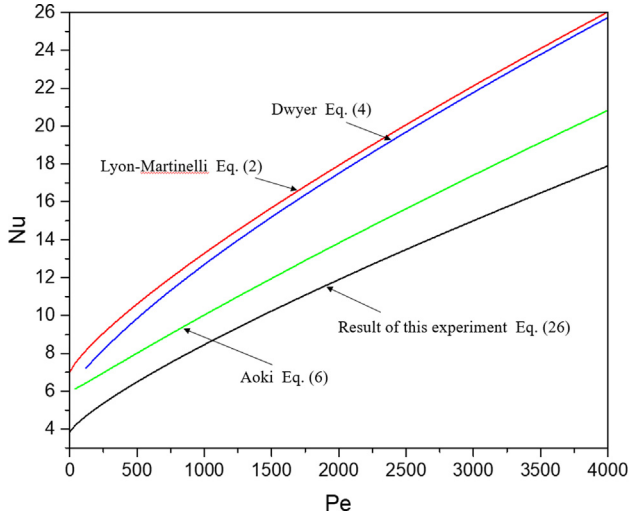


Fig. 12. Comparison of the experimental results and existing theoretical research.

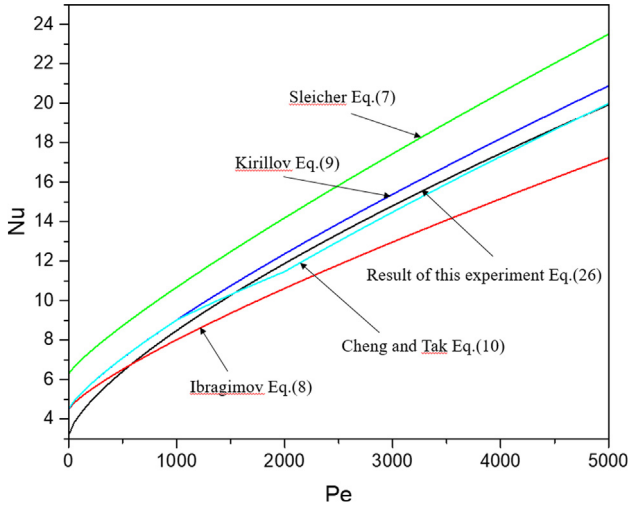


Fig. 13. Comparison of the experimental results.

correlation between the frictional resistance coefficient and *Re* is obtained:

$$f = \frac{0.1581}{\mathrm{Re}^{0.13}} \quad 38000 < \mathrm{Re} < 185000 \tag{27}$$

(2) Key parameters sensitivity study shows that the heat transfer coefficient of LBE in pipes increases with heat flux, *Re* and diameter of the test section, which is different from sodium because of the different physical properties.

(3) Based on the experimental data, a correlation between the *Nu* and *Pe* is fitted as Eq. (28). The correlation is appropriate in the higher *Pe* region and the correlation of Kirillov et al. and Ibragimov et al. is recommended in the low *Pe* region.

$$Nu = 3.8472 + 0.0179Pe^{0.8037} \quad 350 < Pe < 3800 \tag{28}$$

CRediT authorship contribution statement

Yan Zhang: Investigation, Validation. **Chen glong Wang:** Writing - original draft. **Rong Cai:** Formal analysis. **Zhike Lan:** Writing - review & editing. **Yaou Shen:** Project administration. **Dalin Zhang:** Conceptualization. **Wenxi Tian:** Visualization. **Guanghui Su:** Data curation. **Suizheng Qiu:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Financial support for this work was provided by the National Natural Science Foundation of China (No. 11675162) and National Key Research and Development Program of China (No. 2019YFB1901300).

References

[1] J.E. Kelly, Generation IV International Forum: a decade of progress through international cooperation, Prog. Nucl. Energy 77 (2014) 240–246.
[2] O.N.E. Agency, Technology Roadmap Update for Generation IV Nuclear Energy Systems, 2014.
[3] F. Roelofs, A. Gerschenfeld, J. Pacio, N. Forgione, X. Cheng, Thermal Hydraulics Aspects of Liquid Metal Cooled Nuclear Reactors, Woodhead Publishing, 2019.
[4] Y. Zhang, C. Wang, Z. Lan, S. Wei, R. Chen, W. Tian, G. Su, Review of thermal-

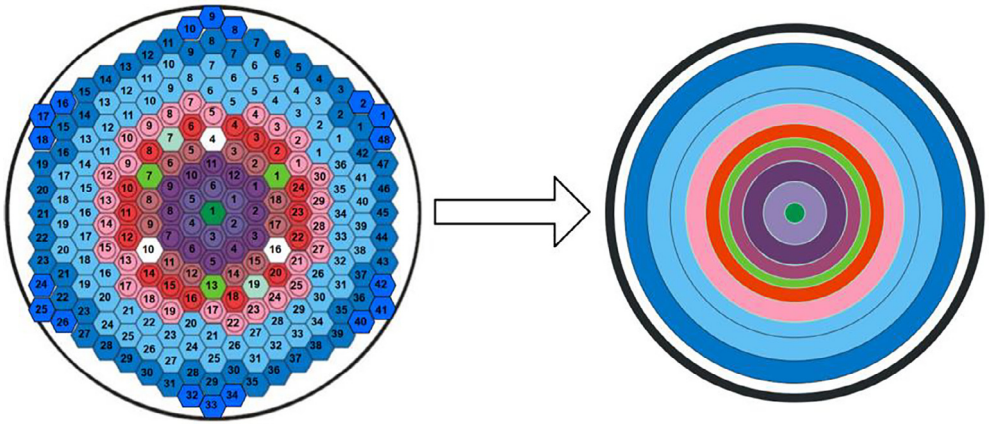


Fig. 4. Schematic diagram of the equivalent circle of the core.

Table 2
Main design parameters of SNCLFR-10.

Parameters	Numerical Value
Thermal power	10 MW
Coolant	LBE
Drive method	Natural circulation
Primary circuit operating pressure	0.1 MPa
Core inlet temperature	533 K
Core outlet temperature	663 K
Total mass flow rate through the core	529.4 kg/s
Circulation height	2 *m*
Horizontal distance between core and heat exchanger	1.8 *m*
Average core flow rate	0.12356 m/s
Active zone height	0.8 *m*
Active zone width	2.142 *m*
Equivalent diameter of fuel zone	1.08 *m*
Main container height	8.8 *m*
Main container width	5.60 *m*
Cold source height	1.52 *m*
Cold source diameter	0.40 *m*

Table 3
Design parameters of the experimental facility under natural circulation.

Parameters	Experimental facility design parameters
Power	1.77 MW
Coolant	LBE
Heating method	electric heating
Cooling method	water cooling
Cold source height	0.76 *m*
Cold source diameter	0.20 *m*
Heat source height	0.40 *m*
Heat source diameter	0.26 *m*
Circulation height	1.00 *m*
Total height of the experimental loop	3.00 *m*
Total width of the experimental loop	2.00 *m*
Pipe diameter	0.20 *m*
Heat source inlet temperature	533.00 K
Heat source outlet temperature	663.00 K
Total mass flow rate through the core	93.59 kg/s
Loop operating pressure	0.10 MPa

3.3.1.2. Thermophysical properties of coolant. LBE is used as the coolant and its FLUENT parameters are input as shown in Table 4 (Chirkin, 1968):

3.3.1.3. Sensitivity analysis of grid. Computational resources should be saved as much as possible, and the computation time should be shortened under the premise of credible results. Therefore, it is necessary to consider the relationships between the total flow and the average temperature at the outlet of the active zone of the core with the number of grids of the model. In particular, for ensuring consistent and credible results, the optimal number of grids has been calculated to optimize the calculation time by setting different amounts of grids. The analysis results are shown in Fig. 6.

As shown in Fig. 6, when the number of the grids in model reaches 9×10^4, the total mass flow rate and the average temperature at the outlet of the active zone of the core tend to be stable; at this time, the effect of increasing the grid number on the calculation results can be neglected.

Thus, the scheme with a grid number of 9×10^4 can be used to obtain accurate calculation results in a shorter period of time.

3.3.1.4. Turbulence model selection and methods. Three CFD numerical simulation methods are commonly used: Reynolds time-averaged method, large eddy simulation method, and direct simulation method (Ozalp and JayaKrishna, 2010). At present, the Reynolds time-averaged method is frequently used for liquid metal cooling systems, and its main turbulence models include $k-\varepsilon$ (Reynolds stress model) model, Rennormalization Group (RNG) $k-\varepsilon$ model, Shear-Stress Transport (SST) $k-\varepsilon$ model. Among them, the RNG $k-\varepsilon$ model exhibits a higher solution accuracy for flow fields with turbulent vortices and large velocity gradients. The reactor coolant would be strongly mixed in the lower chamber, thereby producing more turbulent vortices with large velocity variations in the core; therefore, the RNG $k-\varepsilon$ turbulence model is chosen for the calculation of the flow heat transfer of LBE. Detailed input parameters for Fluent are shown in the following Table 5:

3.3.1.5. Analysis of model results. The numerical simulation analysis is carried out based on FLUENT under the simplified conditions mentioned above, and the results are shown in Fig. 7 and Table 6.

As can be seen from the results, the numerical simulation results of SNCLFR-10 are in good agreement with the design value, and the absolute value of the maximum relative error is less than 8%. Thus, it can be considered that the established SNCLFR-10 CFD model affords optimum certain credibility and accuracy (Table 7).

3.3.2. Numerical simulation of scaled-down device

3.3.2.1. CFD analysis model. Fig. 8 shows the schematic diagram of the two-dimensional model of the scaled-down experimental facility. It is evident that the experimental facility is designed to exhibit a shell structure. Consequently, if the two-dimensional model is selected for numerical simulation, the average flow velocity of

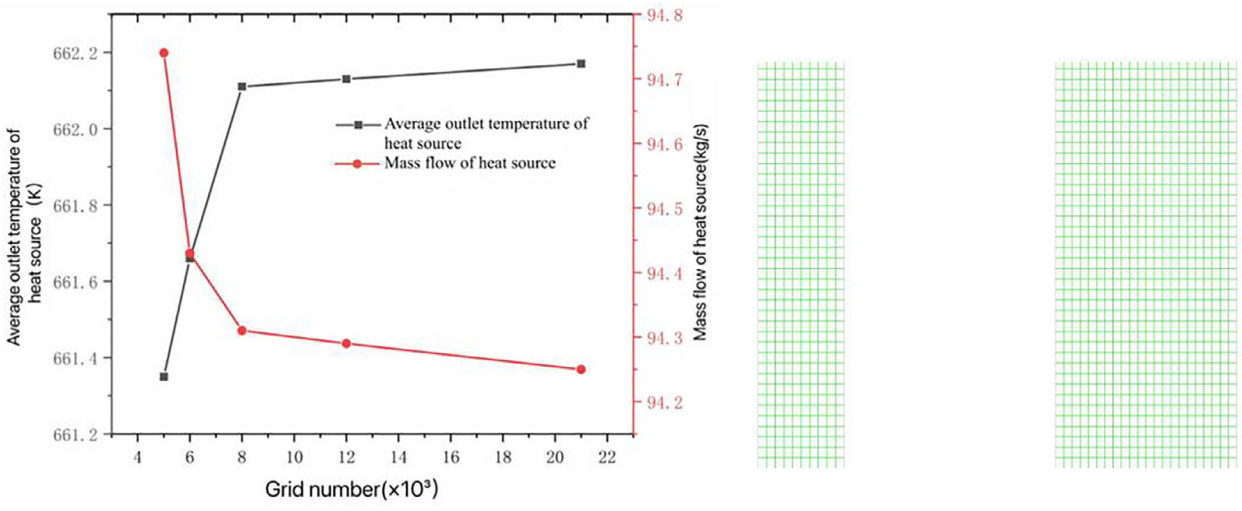


Fig. 9. Grid sensitivity of the scale-down experimental facility.

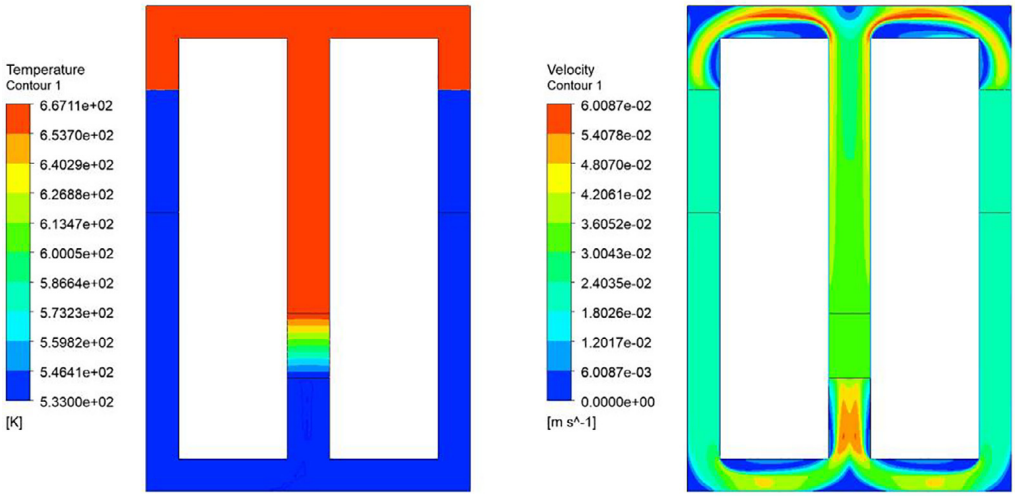


Fig. 10. Temperature distribution of the scaled-down experimental facility.

Table 8
Results of the numerical simulation of the scaled-down experimental device.

Parameters	Design value	CFD result	Relative Error
Inlet temperature	533 K	533 K	0
Outlet temperature	663 K	666.51 K	0.53%
Mass flow rate	93.58 kg/s	91.23 kg/s	−2.51%
Average flow velocity of heat source	0.06178 m/s	0.0646 m/s	4.56%

circulation are compared and analyzed to determine whether SNCLFR-10 and the experimental facility meet the desired proportional relationship, as shown in Table 9.

Table 9 shows that the actual ratios of key thermal–hydraulic parameters of SNCLFR-10 and the experimental facility are in good agreement with the theoretical ratios. Furthermore, the maximum relative error is 5%. Generally speaking, the error of CFD numerical simulation is about 10–15%. Because this model is a two-dimensional model, three-dimensional flow is not considered, so its accuracy cannot be very high, but in engineering, the maximum error is within a reasonable range. Based on these results, it can be concluded that the proportional method is reasonable as well as feasible, and the method can provide theoretical guidance for developing lead-based fast reactors with natural circulation.

Table 9
Numerical simulation results of SNCLFR-10 and experimental facility.

Parameters	SNCLFR-10	Experimental facility	Actual ratio	Theoretical ratio	Relative Error
Heat source inlet temperature	533 K	533 K	1.00	1.00	0.00%
Heat source outlet temperature	662.07 K	666.51 K	1.01	1.00	1.00%
Temperature difference between inlet and outlet	129.07 K	133.51 K	1.03	1.00	3.00%
Mass flow rate	526.8 kg/s	91.23 kg/s	5.59	5.66	0.54%
Natural circulation Drive	222675 Pa	106186 Pa	2.10	2.00	5.00%
Average flow velocity of heat source	0.1244 m/s	0.0646 m/s	1.93	2.00	−3.50%

表 2　实验装置主要结构参数

Tab. 2　Main Structural Parameters of Experimental Facilities

装置	参数	原型	1/2模型	1/4模型
热源	有效加热段高度/mm	3660	1830	915
	热源总高度/mm	4634	2804	1889
	子通道当量直径/mm	11.8		
	电加热管规格/mm	Φ21×2		
	承压壳规格/mm	Φ255×32.5		
热阱	筒体高度/mm	12815	6411	3208
	筒体规格/mm	Φ114.3×8		
	传热管高度/m	10605	5282	2611
	传热管规格/mm	Φ19×2		
	传热管数目/根	2		
一回路管道	冷热芯形位差/m	16	8	4
	冷管直径/mm	17.0		
	冷管长度/m	6	3	1.5
	热管直径/mm	18.0		
	热管长度/m	4.9	2.45	1.225

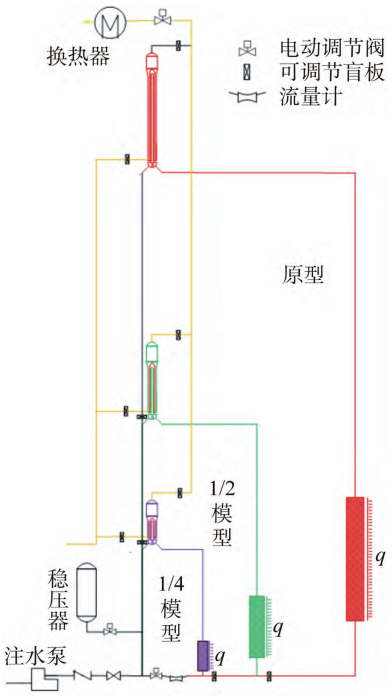


图 1　实验装置流程图

Fig. 1　Flow Chart of Experimental Facilities

口压差及各管段压差均进行了测量，一回路流量计布置在热源的入口管段。热源功率通过直流电源加载，同时测量其电流和电压信号。所有测量信号经 NI 公司的采集系统进行模数转换后送计算机系统进行显示、运算和存储。其中温度的测量精度为±1℃，压力和压差的测量精度为 0.1%，电流的测量精度为 0.1%，电压的测量精度为 0.1%。

1.3　实验装置的阻力参数标定

确保缩比后实验装置的阻力系数与原型保持一致，是重要的模化准则之一。在原型中，各设备、管道的结构参数固定，而阻力系数是热工参数的相关量，模化后的实验装置无法保证每一个工况下的阻力系数满足模化准则，事实上，缩比模型仅在单相下进行阻力标定试验，仅确保在单相工况下的阻力系数不严重失真。对于依据比例模化分析结果设计的各尺度模化装置，先在单相环境下开展模化装置的单相阻力标定实验，通过阀门、节流件等来调节模化装置的整体阻力系数和关键设备的阻力系数[11]。图 2 以原型为例，给出了原型下降段阻力系数在不同压力（P）下的变化曲线。可见在单相环境下，阻力参数随 P 的变化很小。在一定的流速范围内，阻力参数可实现较高精度的模化标定。开展两相自然循环实验工况时，不再调节阻力件，因此，两相环境下的阻力系数模化失真要大于单相环境。

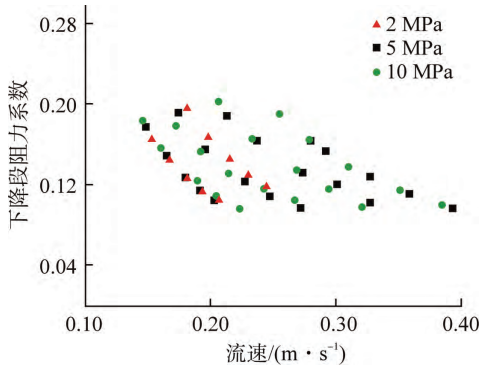


图 2　不同压力下的原型下降段阻力系数

Fig. 2　Resistance Coefficient of Prototype Descent Pipe under Different Pressure

图 3 为自然循环系统中上升段、下降段、热源、热阱等主要流道的阻力参数标定。由图 2 可知各设备及管道的阻力系数受压力影响较小，这主要是因为在固定结构参数下，流体粘度是阻力系数的主要影响因素，而粘度受压力影响较小。因此，图 3 中不再对不同压力下的数据进行区分。在单相环境下，上升段、下降段、热源、热阱的

(C)1994-2023 China Academic Journal Electronic Publishing House. All rights reserved.　http://www.cnki.net

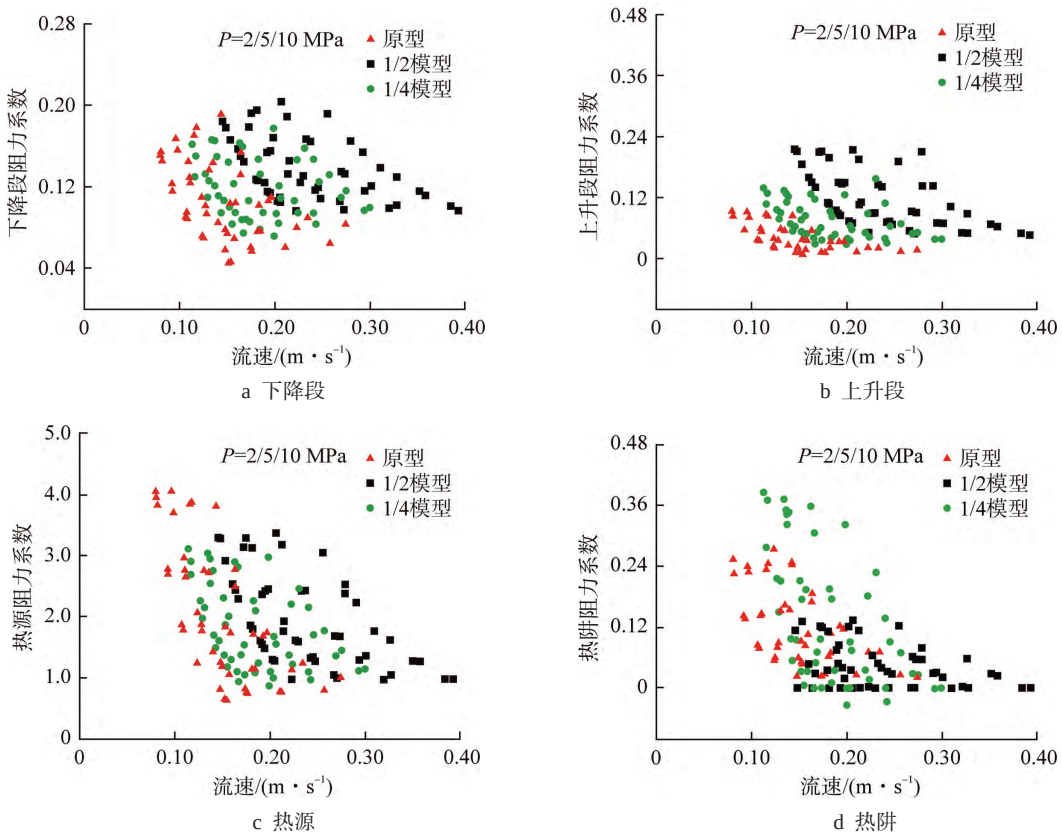


图 3 系统各主要设备和管道阻力标定

Fig. 3 Resistance Calibration for Main Equipment and Pipeline

阻力系数随流速变化较小，原型、1/2 模型、1/4 模型的各区域阻力系数均在同一水平，但并非完全相等，这主要是由设备加工精度和设计偏差导致的。实验过程中，通过调节布置在热源入口段的电动调节阀来进行总体阻力系数的调整，以满足单相环境下的模化装置阻力标定要求，此时原型一回路阀门开度为 100%，1/2 模型一回路阀门开度为 19%，1/4 模型一回路阀门开度为 13%。

2 自然循环特性曲线模化失真评价

2.1 自然循环流量-热功率特性曲线失真评价

根据比例模化推导准则，原型、1/2 模型和 1/4 模型的自然循环流量及加热功率应遵循以下准则：

$$\begin{cases} \dfrac{Q_{\mathrm{M},1/2}}{Q_{\mathrm{P}}}=\dfrac{\sqrt{2}}{2},\dfrac{Q_{\mathrm{M},1/4}}{Q_{\mathrm{P}}}=\dfrac{1}{2} \\ \dfrac{G_{\mathrm{M},1/2}}{G_{\mathrm{P}}}=\dfrac{\sqrt{2}}{2},\dfrac{G_{\mathrm{M},1/4}}{G_{\mathrm{P}}}=\dfrac{1}{2} \end{cases} \tag{1}$$

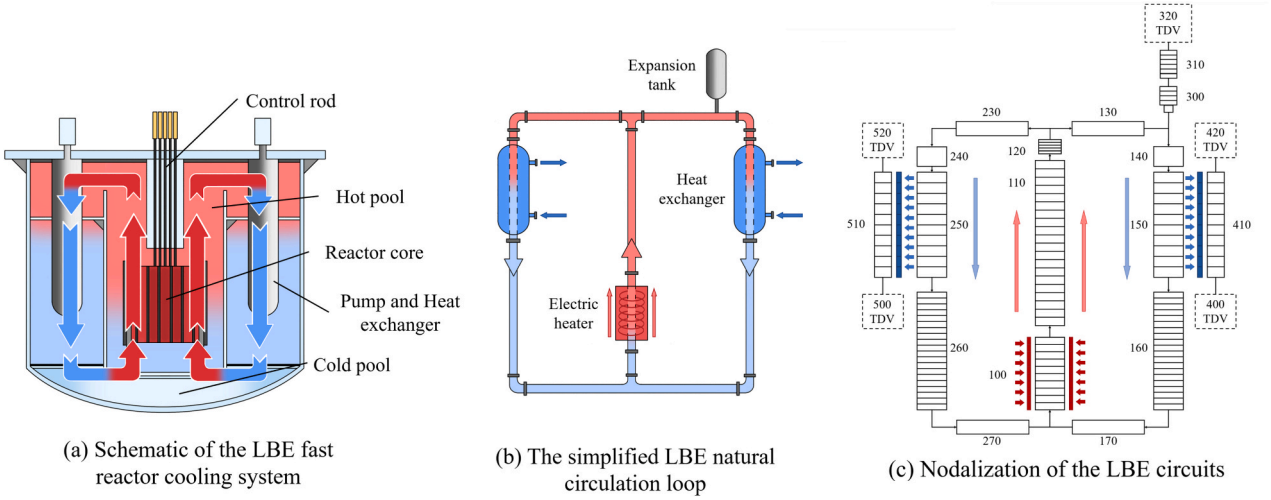
图 4 为稳态、单相工况下自然循环流量与加热功率间特性曲线，其中图 4a 为各装置的流量-加热功率（*G*-*Q*）特性曲线直接对比，图 4b 中将 1/2 模型和 1/4 模型的 *G*-*Q* 特性曲线根据模化准则式（1）反向推导至原型参数，并与原型实验装置的 G_{P}-Q_{P} 特性曲线进行比较。由图 3 可见，单相自然循环流量受压力的影响较小。原型、1/2 模型、1/4 模型的自然循环流量-加热功率特性曲线均呈现幂函数形式，拟合函数的均方差为 0.98，表明在单相环境下，降高度比例模化后的模型装置能够复现原型中的 G_{P}-Q_{P} 特性曲线。

图 5 为两相工况下自然循环流量与加热功率的特性曲线，在两相环境下，自然循环流量的实测值有小幅度的波动，故取时均值。在两相环境下，自然循环流量与加热功率呈近似线性关系，但降高度模化后的自然循环流量的失真度明显大于单相环境下的失真度，这主要是因为两相自然循环的阻力系数模化失真比单相自然循环时更大。

2.2 雷诺数-格拉晓夫数特性曲线失真评价

格拉晓夫数（*Gr*）在一定程度上反应了自然

(C)1994-2023 China Academic Journal Electronic Publishing House. All rights reserved. http://www.cnki.net



(a) Schematic of the LBE fast reactor cooling system

(b) The simplified LBE natural circulation loop

(c) Nodalization of the LBE circuits

Fig. 1. Schematic diagram of LBE natural circulation and its simplified loop.

Table 1
Design parameters of the prototype case under LBE natural circulation.

Parameters	Value
Coolant	LBE
Pressure (MPa)	0.1
Total height (m)	2.1
Total width (m)	2.04
Pipe inner diameter (mm)	30
Heating section length (m)	0.4
Cooling section length (m)	0.65
Height between cooling and heating sections (m)	1.375
Initial power (KW)	0–22

2. Physical model

2.1. LBE single-phase natural circulation loop

Fig. 1a shows the simplified phenomenon of the natural circulation in the traditional lead-cooled reactor. When the main pump fails or the reactor is designed to operate in a passive mode, the heat of the core is removed by the natural circulation. Experimental loops were often used to simulate the LBE flow inside the reactors (Ma et al., 2007; Agostini et al., 2002; Coccoluto et al., 2011). The simplification principles of the model are as follows:

(a) LBE flows along the uniform and adiabatic pipes within the experimental loop and the flow non-uniformity at certain pipe cross-sections caused by the measurement instruments is ignored.

(b) The Boussinesq hypothesis was applied to the facility (Chen et al., 2012).

(c) The core was regarded as a point heat source and the power was calculated based on the scaling methods.

A symmetrical loop was obtained as the prototype case for scaling analysis. The loop is illustrated in Fig. 1b. After flowing through the electric heater that simulates the reactor core, the LBE reached the top of the loop and it was divided into two parts to enter the symmetrical heat exchangers. The natural circulation within the loop was driven by the density variations resulting from the temperature difference at the centers of the cooling and heating sections. The main parameters of the prototype loop are shown in Table. 1.

2.2. Node model of the LBE loop

The LBE loop shown in Fig. 1b was simulated using the Relap5/Mod4.0 code. After the relevant improvements, this code can conduct the thermal–hydraulic calculations for the lead–bismuth fluids (Kumari and Khanna, 2017). The node diagram of the loop is shown in Fig. 1c.

The hot section of the loop was simulated by Pipe 100, 110, and 120, among which Pipe 100 was used to simulate the reactor core. Similarly, Pipe 140-160 and Pipe 240-260 were used to simulate the symmetrical cold sections. The heat of the LBE flow was removed by the heat exchangers represented by Pipe 150 and 250. To maintain pressure at the initial value, pipe structures 300 and 310 and time-dependent control volumes TDV320 were used to simulate the expansion tank for the LBE loop.

Meanwhile, the number of nodes has been chosen to obtain a trade-off between the computational time and the correct representation of the physical phenomena (Urbonas et al., 2003). The heating and cooling pipe sections were analyzed, and the number of their nodes was set to 10 as shown in Fig. 1c.

3. Scaling analysis of the LBE natural circulation loop

3.1. Scaling analysis methods

H2TS and DSS are interesting scaling analysis methods (Zuber, 1991; Reyes, 2015) and have been discussed many times in previous studies (Li et al., 2018; Xu et al., 2022). For the H2TS analysis method, the relevant criteria array is obtained by dimensionless control equations. The scale condition of the criterion array is given to obtain the final scaling criterion. Meanwhile, the DSS is different. This theory describes the physical processes as geometric objects in a normalized coordinate system and the control equations are treated by the overall method (Reyes, 2015). Finally, the scaling criterion is obtained.

However, the previous analyses were all based on the water loops. To expand their application in the field of LBE natural circulation and guide the design of the scaled-down experimental facilities, the above theories were applied to the loop shown in Fig. 1c. The principles of different scaling methods and the treatment of the conservation equations were comprehensively described in the previous studies (Li et al., 2018). Only the key steps of each method will be described and the detailed scaling criterion will be shown here.

First, the integral momentum and energy equations of the loop are given by:

$$\frac{l}{A}\frac{dw}{dt} = \beta_s g \rho H \Delta T - \frac{w^2}{2\rho A^2}\left(\frac{fl}{d} + k\right) \tag{1}$$

$$\rho c_p \left(\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial z}\right) = \frac{q\xi}{A} \tag{2}$$

Considering the simplification principle (a) for the model, the energy

Table 2
Scaling number group based on four scaling methods.

Method serial number	Method 1	Method 2	Method 3	Method 4
Description	H2TS	Zhao's method References (Zhao et al., 2020)	DSS Identity	DSS Dilation $\lambda = 1.25$
d_R	$l_R^{\frac{7}{10}}$	l_R	$l_R^{\frac{3}{5}}$	$1.367 l_R^{\frac{3}{5}}$
u_R	$l_R^{\frac{1}{2}}$	$l_R^{\frac{1}{2}}$	l_R	$1.25 l_R$
w_R	$l_R^{\frac{19}{10}}$	$l_R^{\frac{5}{2}}$	$l_R^{\frac{11}{5}}$	$2.335 l_R^{\frac{11}{5}}$
Q_R	$l_R^{\frac{19}{10}}$	$l_R^{\frac{5}{2}}$	$l_R^{\frac{16}{5}}$	$2.919 l_R^{\frac{16}{5}}$
ΔT_R	1	1	l_R	$1.25 l_R$

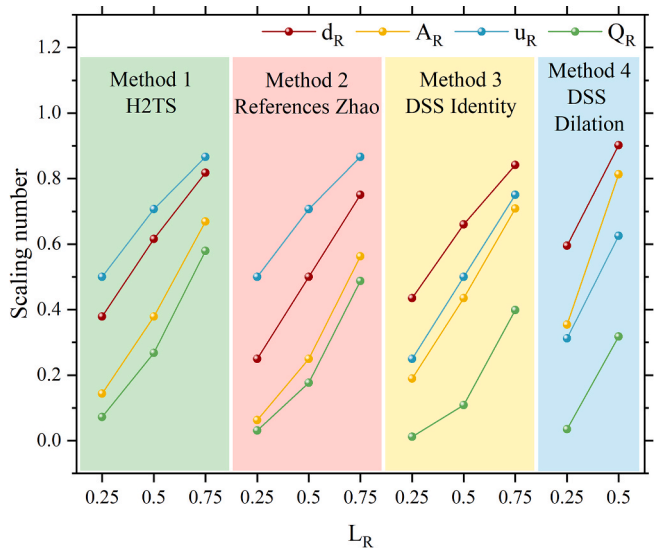


Fig. 2. Scaling number curve with length ratio.

equation can be simplified to:

$$\rho c_p\left(\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial z}\right)=\frac{4q}{d} \tag{3}$$

Eqs. (1) and (3) have the dimensionless form, the superscript "+" indicates the dimensionless form of the parameters:

$$\frac{du^+}{dt^+}=\Delta T^+\frac{\beta_s g H\Delta T_0}{u_0^2}-\frac{1}{2}\left(\frac{fl}{d}+k\right)(u^2)^+ \tag{4}$$

$$\frac{\partial T^+}{\partial t^+}+\frac{\partial T^+}{\partial z^+}=\frac{4ql}{\rho c_p u_0 d\Delta T_0} \tag{5}$$

The dimensionless numbers, including the Richardson number, the Friction number and the heat source number can be obtained through the above equations:

$$\prod_{Ri}=\frac{\beta_s g H\Delta T_0}{u_0^2} \tag{6}$$

$$\prod_{Fri}=\frac{fl}{d}+k \tag{7}$$

$$\prod_{Qs}=\frac{4ql}{\rho c_p u_0 d\Delta T_0} \tag{8}$$

To make the above equations applicable to both the prototype and the scaled-down cases, the following scaling criteria are defined as follows:

$$\prod_{Ri,R}=\left(\frac{\beta_s g H\Delta T_0}{u_0^2}\right)_R=1 \tag{9}$$

$$\prod_{Fri,R}=\left(\frac{fl}{d}+k\right)_R=1 \tag{10}$$

$$\prod_{Qs,R}=\left(\frac{4ql}{\rho c_p u_0 d\Delta T_0}\right)_R=1 \tag{11}$$

The detailed criterion numbers are simultaneously satisfy Eqs. (6) to (8). Substituting $w_R = u_R A_R$ and $Q_R = q_R d_R l_R$, thus:

$$d_R=l_R^{\frac{7}{10}}, u_R=l_R^{\frac{1}{2}}, w_R=l_R^{\frac{19}{10}}, Q_R=l_R^{\frac{19}{10}}, \Delta T_R=1 \tag{12}$$

For the power of the core we have:

In the relevant previous studies, Zhao et al. (Zhao et al., 2020, 2022) discussed a steady case of the LBE natural circulation using a similar method. To comprehensively evaluate the different methods, the scaling criterion obtained from the relevant studies is presented here and will be compared in the following discussion (Zhao et al., 2020).

$$d_R=l_R, u_R=l_R^{\frac{1}{2}}, w_R=l_R^{\frac{5}{2}}, Q_R=l_R^{\frac{5}{2}}, \Delta T_R=1 \tag{13}$$

Equation (10) shows that Zhao's method has the derivation results similar to those of the H2TS method. The theoretical ratio of the flow velocity and the temperature difference is the same as that in Equation (9). The difference is that the diameter ratio is identical to the length ratio and the two methods have different power ratios of the scaled cases.

The derivation of the DSS method is as follows:

For Eq. (1), we have:

$$\omega_w^w=\frac{-\frac{w^2}{2\rho Al}\left(\frac{fl}{d}+k\right)}{w_0}=-\frac{w_0}{2\rho Al}\left(\frac{fl}{d}+k\right)(w^+)^2 \tag{14}$$

$$\omega_T^w=\frac{\beta_s g\rho H\Delta TA}{w_0 l} \tag{15}$$

According to the DSS theory, ω is defined as the normalized sum of agents-of-change. Here, it denotes the dimensionless responses of the agents-of-change on the mass flow rate (Reyes, 2015; Li et al., 2018). This parameter of the scaled model and prototype are ω_M and ω_P. We define: $\omega_R = \omega_M/\omega_P$.

Thus:

$$\omega_{w,R}^w=\frac{w_{0,R}}{(Al)_R}\left(\frac{fl}{d}+k\right)_R\left[\frac{w_M^+(t_M)}{w_P^+(t_P)}\right]^2 \tag{16}$$

$$\omega_{T,R}^w=\left(\frac{HT_0}{lu_0}\right)_R\left[\frac{T_M^+(t_M)}{T_P^+(t_P)}\right] \tag{17}$$

DSS scaling criteria are as follows:

$$\lambda_{w,A}^w=\tau_{S,R}^w\omega_{w,R}^w, \lambda_{T,A}^w=\tau_{S,R}^w\omega_{T,R}^w \tag{18}$$

$$\tau_{S,R}=\frac{\lambda_A}{\lambda_B} \tag{19}$$

In the DSS theory, λ_A and λ_B represent the strain parameters on different coordinates. They determine the specific transformation methods. According to the 2–2 Affine transformation criteria (Reyes, 2015):

$$\beta_R=\lambda_A, \omega_R=\lambda_B \tag{20}$$

τ_S is the process time interval of the DSS transformation (Li et al., 2018). Meanwhile, according to the method:

through steady-state 1D analysis and 3D CFD studies. Borgohain et al. [11] investigated natural circulation in HANS test loop and indicated that natural circulation capability depended on loop structure parameters.

An engineering project to develop ADS and lead-based fast reactors technology was launched in China in 2011, with the CLEAR (China lead-based reactor) being the reference reactor for ADS and fast reactor. It is performed by a number of institutions led by Institute of Nuclear Energy Safety Technology, Chinese Academy of Sciences (INEST). To support the design and construction of CLEAR-I, an LBE rectangular loop named KYLIN-II Thermal Hydraulic (KYLIN-II TH) natural circulation test loop was constructed on the basis of available experience on lead–lithium technology [12], including the fusion reactor design [13–15], R&D on blanket [16, 17], R&D on fusion materials and technologies [18–20]. To verify the capability of natural circulation in KYLIN-II TH natural circulation test loop, it is necessary to analyze flow characteristics of natural circulation. One of the goals of KYLIN-II facility is to carry out natural circulation experiments under different conditions, aiming at, among others, supporting the verification and validation of numerical codes.

In this paper, steady-state experiments on natural circulation with different heating powers are performed in the loop. The results are compared with theoretical analysis and CFD simulation results.

2 KYLIN-II TH natural circulation loop

The KYLIN-II TH natural circulation loop is an LBE rectangular loop as depicted in Fig. 1. The loop basically consists of two vertical pipes working as riser and downcomer, respectively, connected by two horizontal branches. In the bottom part of the riser, a heat source (HS) of 24.0 kW is installed, while the upper part of the downcomer is connected to a heat exchanger (HX). An expansion vessel is installed on the loop top, coaxially to the riser. By a valve installed in the cold part of the loop, the singular pressure losses of the system are modified in the tests. Table 1 shows the main parameters of the loop.

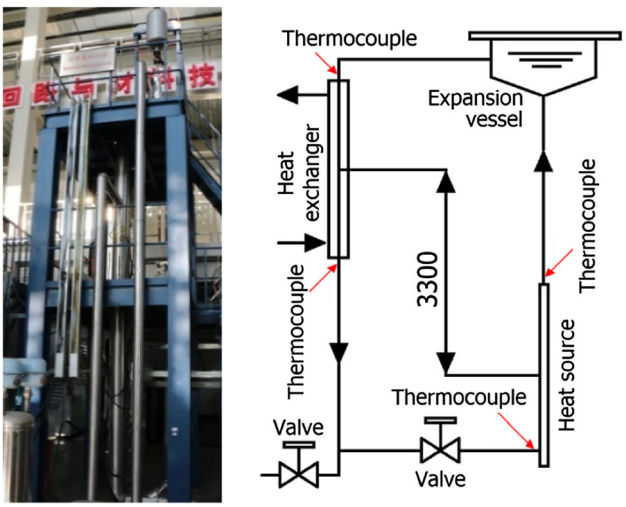


Fig. 1 Layout of the KYLIN-II TH natural circulation loop

The HS is arranged at the bottom of the riser containing an electrical rod simulator, which is concentrically inserted in a 54-mm-ID tube made of 316L stainless steel. The inner diameter of HS is larger than the main pipe of the loop, for the average velocity in the main pipe, it is estimated 0.12–0.25 m s^{-1}, while, and it is about 0.03–0.06 m s^{-1} for the velocity inside the HS. Connected coaxially to the top of the riser, the expansion vessel provides a suitable volume to allow the LBE expansion when it is heated to hundreds of degrees in the test. To avoid the LBE being oxidized, the free level of LBE is covered by argon gas with a slight overpressure of tens of kPa. Considering the heat exchange capability without introducing extra pressure loss, a 'tube-in-tube,' counter-flow-type heat exchanger is adopted. High-temperature heat-conducting oil is used as the secondary fluid. Four immersion *K*-type thermocouples are installed at the both ends of the HS and HX to measure the loop temperature.

3 Theoretical analysis

3.1 Theoretical analysis model for natural circulation

A one-dimensional theoretical model is adopted for analyzing natural circulation in the KYLIN-II TH natural circulation loop. For calculating the LBE flow rate through

Table 1 Main parameters of the KYLIN-II TH natural circulation loop

Parameters	Values
Total height (m)	5
Length of HS (mm)	1800
Pipe inner diameter (mm)	26
HS inner diameter (mm)	54
Electrical rod diameter (mm)	22
Active heating length (mm)	800
Operating temperature (°C)	150–500
Operating pressure (atm)	1–1.5
Input power (kW)	0–24
Maximum wall heat flux (W cm^{-2})	43.4

coefficient declines with the mass flow rate, mainly because the K_{pipe} declines with increasing Reynolds number.

By solving Eq. (5), one can estimate the equilibrium mass flow rate of natural circulation in the loop at different input powers. The main thermal-hydraulic parameters at different power levels are summarized in Table 3.

Table 3 Thermal-hydraulic performance matrix in natural circulation

P (kW)	$\dot{m}$ (kg s^{-1})	ΔT (°C)	v (m s^{-1})
4	0.657	41.4	0.119
8	0.834	65.3	0.151
12	0.957	85.3	0.174
16	1.058	102.9	0.192
20	1.143	119.1	0.208
24	1.215	134.4	0.221

4 The CFD simulation

4.1 Description of CFD modeling

CFD simulations were carried to evaluate the KYLIN-II natural circulation loop, together with the expansion tank. The geometry model of expansion tank, heating rod and HX was adopted as shown in Fig. 2. The valve at the bottom was replaced by a tube. The 3D geometry was modeled in ICEM CFD (the Integrated Computer Engineering and Manufacturing code for Computational Fluid Dynamics) with dimensions of the pipes and other key component described in Sect. 2. The heating volume did not consist of solid volume; only a surface source on the pin wall was adopted. A commercial code FLUNET was employed to simulate the steady state of natural circulation at different heating powers.

The simulation model is RNG κ-ε model with standard wall treatment [11]. The heating powers of 4.3, 7.6, 11.2, 15.3 and 17.9 kW were decided on experimental basis. Correspondingly, the heat fluxes of 5.15, 9.20, 13.47, 18.40 and 21.52 W cm^{-2} were imposed on the rod bundle surface. Regarding the HX, a heat transfer coefficient of 4.89 W cm^{-2} K^{-1} and a free stream temperature of 200 °C were utilized on the oil side. After a sensitivity analysis on the unstructured mesh, the grid numbers were defined as 1.3×10^6 °C.

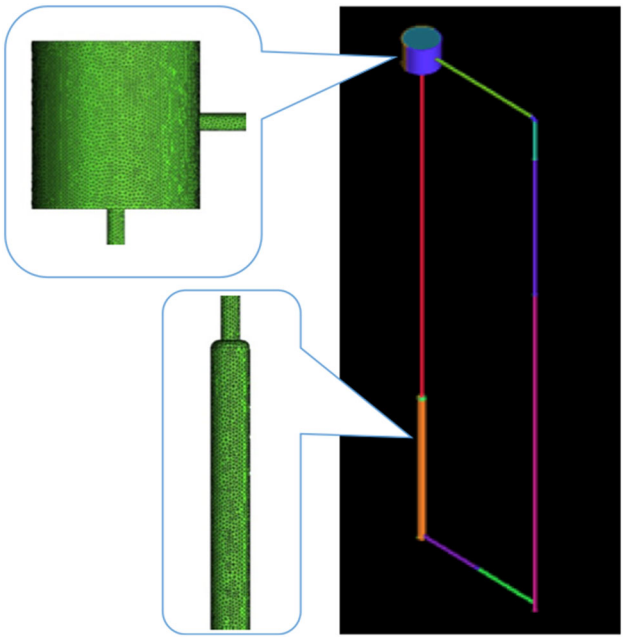


Fig. 2 3D geometry model and unstructured grid mesh of the loop

4.2 CFD simulation results of the steady-state experiments

The CFD results at different input powers were obtained using FLUENT. The temperature and velocity contours at heating power of 17.9 kW are shown in Fig. 3. The maximum temperature appeared on upper end of the heating rod (Fig. 3a). The loop was assumed adiabatic, without heat losses, so the outlet temperature of HX matches with the inlet temperature of HS. The HS outlet temperature was 334.4 °C, and the temperature difference through the HS was 101.5 °C. From Fig. 3b, the velocity through all pipes was almost constant, hence a steady state.

Table 2 Reynolds numbers, resistance coefficients and overall pressure loss as a function of the LBE mass flow rate

$\dot{m}$ (kg s^{-1})	Re_{tube} ($\times 10^3$)	Re_{HS} ($\times 10^3$)	K_{HS}	K_{EV}	K_{pipe}	K_{total}	ΔP_{fr} (Pa)
0.25	5.56	1.90	2.9	2.5	19.1	24.5	261.9
0.5	11.10	3.81	2.8	2.5	18.9	24.2	1034.8
0.75	16.70	5.71	2.8	2.5	18.5	23.8	2289.9
1	22.30	7.62	2.7	2.5	18.1	23.3	3985.4
1.25	27.80	9.52	2.6	2.5	17.7	22.8	6093.6